PHARMACOKINETICS 1:
INTRODUCTION, ABSORPTION & DISTRIBUTION

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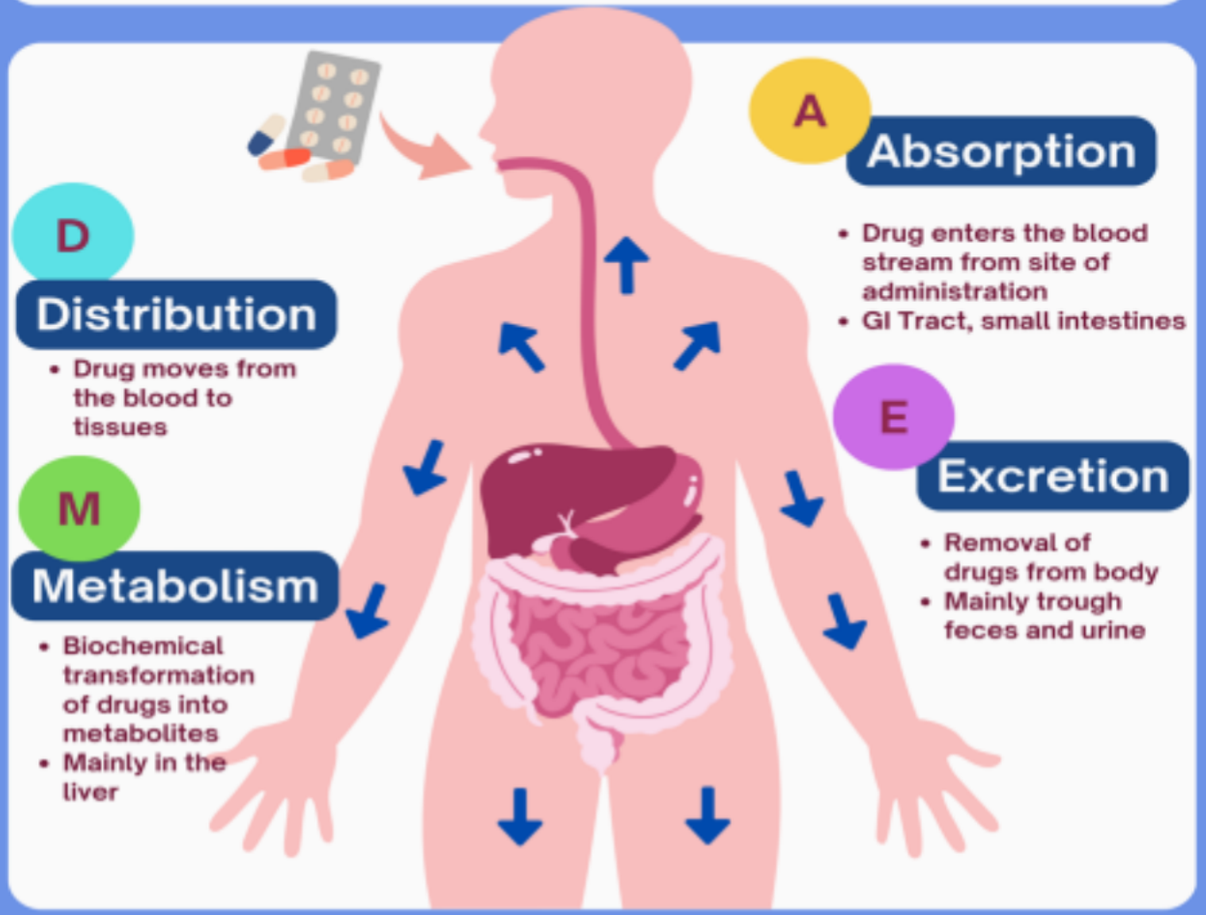
Universiti Malaysia Sarawak

Pharmacokinetics

What happens to drug after you take it?

Four main processes:

Absorption, Distribution, Metabolism, Excretion



A Absorption

- Drug enters the blood stream from site of administration
- GI Tract, small intestines

D Distribution

- Drug moves from the blood to tissues

M Metabolism

- Biochemical transformation of drugs into metabolites
- Mainly in the liver

E Excretion

- Removal of drugs from body
- Mainly trough feces and urine

Applications of pharmacokinetics:

- Drug development
- Dosage adjustments
- Predict drug interactions

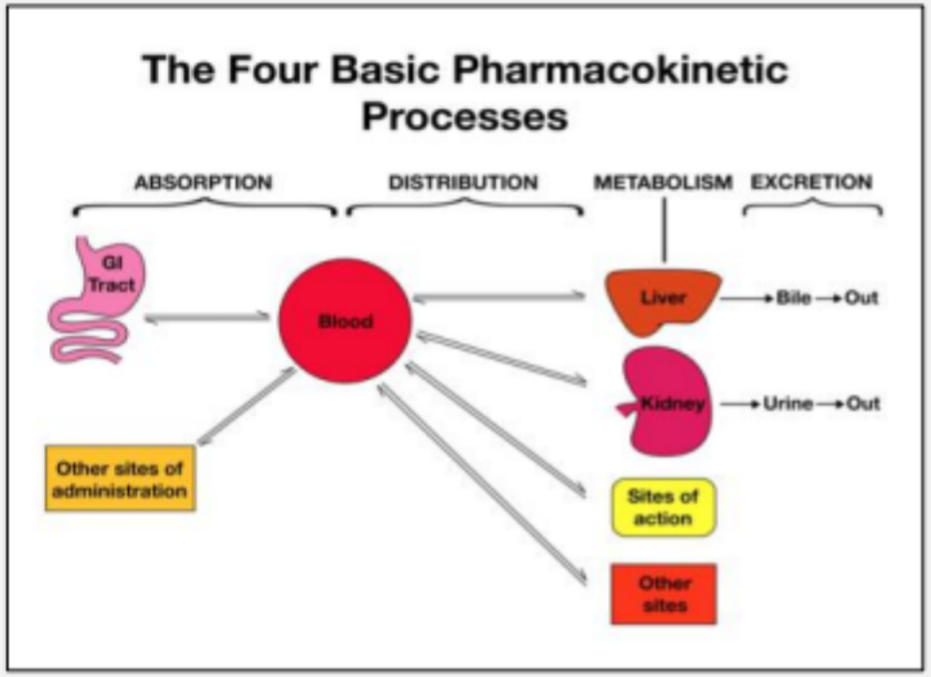
PHARMACO - KINETICS

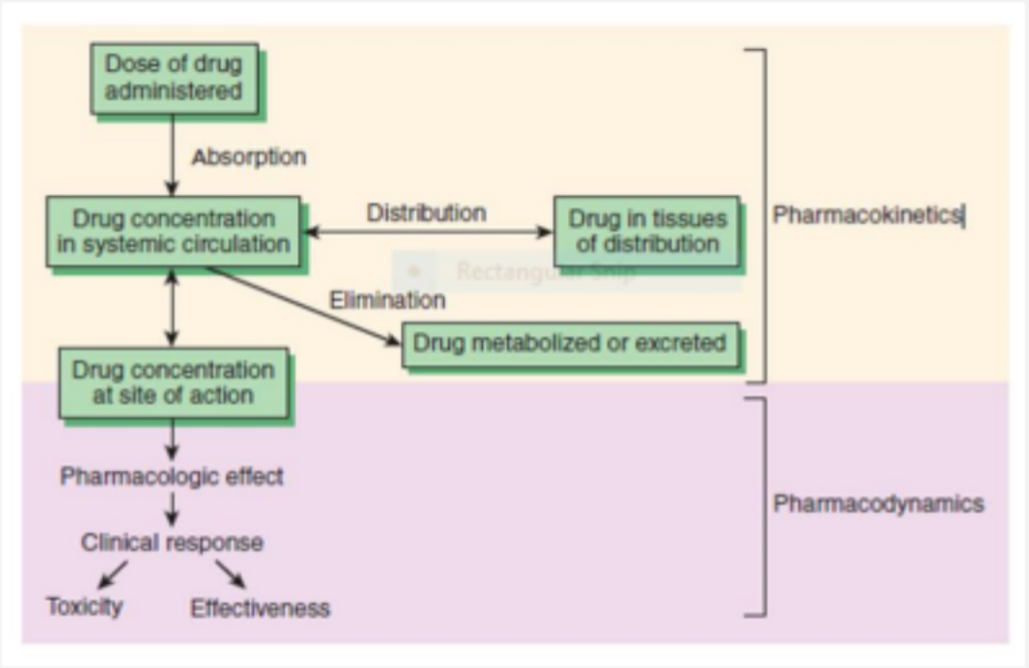
drugs *movement*

Movement of drugs in the body
@
What does the body do to the drug?

A D M E

- Absorption
- Distribution
- Metabolism
- Excretion





OBJECTIVES

- How the onset of drugs and magnitude of drug action are related to the rate and extent of absorption.
- Factors affecting the drug absorption from gastrointestinal tract.
- How ionization of drug influences the absorption.
- First pass effect of drug and how to overcome it.
- Factors influencing the bioavailability of drug.
- Determinants of drug distribution.
- How plasma protein binding influences drug actions

ABSORPTION

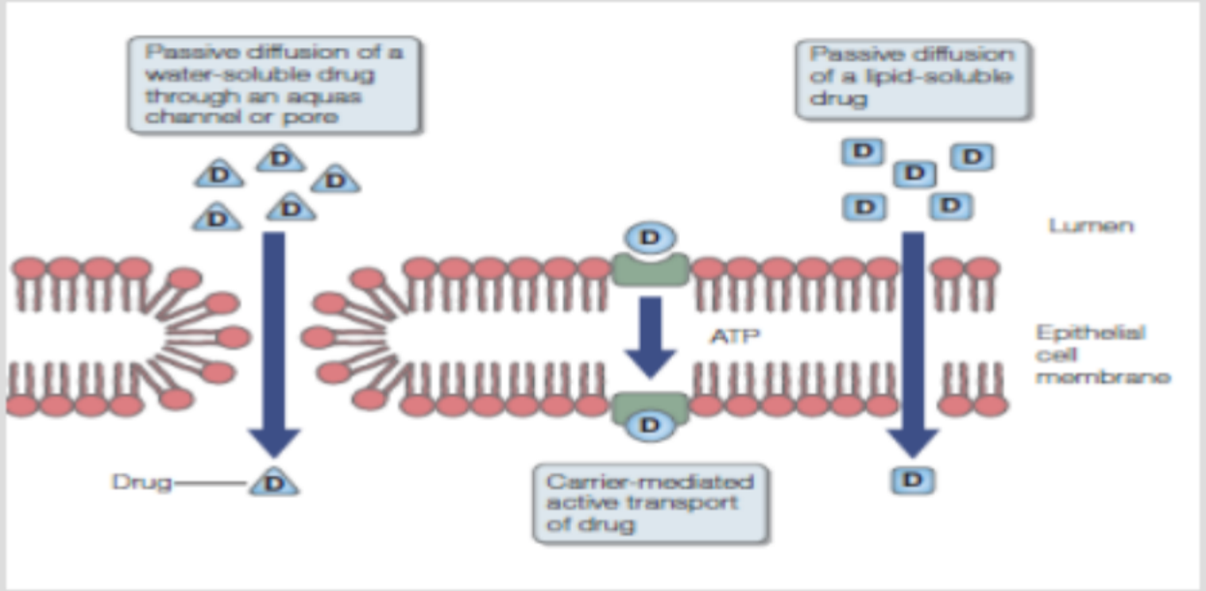
- Definition: Movement of drugs from its site of administration across the cell membrane into the blood stream.
- Drug has to be in a *solution form* before crossing the cell membrane. Therefore, any dosage form except for IV injection, oral solutions, enema, eye drops has disintegrate / dissolve before absorption.

溶物

Modes of drug absorption from the gut

- Passive diffusion:
  - ✓ High concentration to low concentration
  - ✓ Water soluble drugs, lipid soluble drugs
  - ✓ Small molecules

- Active transport:
  - ✓ Polar molecules
  - ✓ Require energy (ATP)
  - ✓ Carrier mediated



ABSORPTION

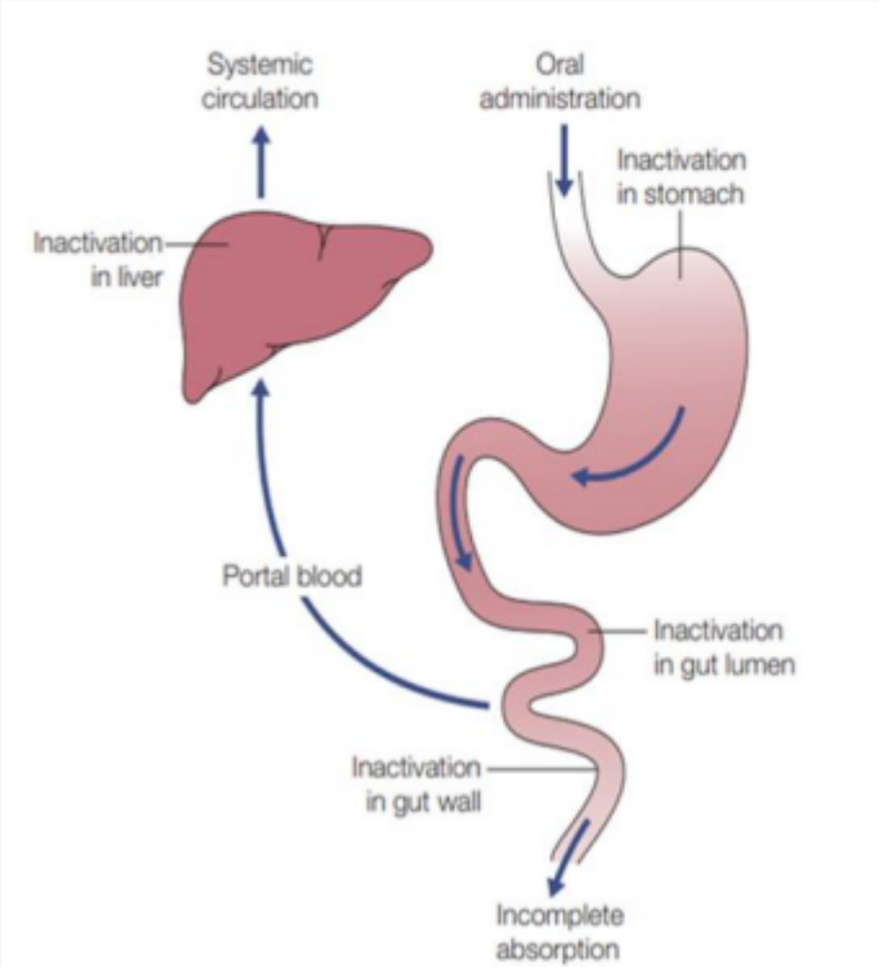


Figure 4.1: Drug bioavailability following oral administration may be incomplete for several reasons.

Besides passing through the intestinal mucosa, a drug that is **orally administered** needs to pass through the liver first before accessing the systemic circulation.

This is called the FIRST-PASS EFFECT. / presystemic metabolism

Large proportion of a drug is chemically changed into inactive metabolites in the liver, then a much smaller amount of drug will pass into the circulation.

BIOAVAILABILITY

The extent of drug absorption (how much drug reaches circulation)

- Bioavailability, F - fraction of unchanged drug reaching the systemic circulation

The FIRST-PASS EFFECT can be a problem:

- Dosages of oral drugs may need to be increased
- Drug effect may be slow which may not be suitable in emergency / urgent cases (e.g. seizures, high fever, acute angina)
- You only need a local effect (e.g. rash, hemorrhoids)

因为多 high dosage, 所以will lead to systemic side effect and toxicity.

Extent of Absorption (how MUCH the drug reaches the circulation?):

- **Bioavailability (F)** - *fraction of unchanged drug reaching the systemic circulation following administration by any route*
- To get 100% bioavailability in bloodstream, you need to give it intravenously (IV).

How to overcome or bypass first pass effect?

1. Use other route of administration or dosage forms
2. Prodrugs such as enalapril gets metabolized to its active form to enaprilat in the liver.

ACE inhibitor prescription medication
- hypertension
- heart failure
- left ventricular dysfunction
- kidney protection

- a prodrug that turn into enalaprilat in body
- block enzyme that produce AT II (substance that narrow blood vessels)

side effect:
- persistent dry cough
- dizziness
- headache, nausea, fatigue
- hyperkalemia / change in kidney function

Box 2.2

Drug Routes and First-Pass Effects

First-Pass Routes

Hepatic arterial
Oral
Portal venous
Rectal[a]

Non–First-Pass Routes

Aural (instilled into the ear)
Buccal 脸颊
Inhaled
Intraarterial
Intramuscular
Intranasal
Intraocular
Intravaginal
Intravenous
Subcutaneous
Sublingual 舌下
Transdermal 透皮

[a]Leads to both first-pass and non–first-pass effects.

Rate of Absorption (how FAST the drug is absorbed into the circulation)

Rate of Absorption can be described by

First Order Kinetics:

Drug is absorbed at a ***constant fraction***

Rate of absorption is proportional to the concentration of drug at the site of administration

e.g. 100mg → 50mg → 25mg → 12.5mg → etc….

[drug is absorbed at a fraction of half at each time]

Example: most drugs

Zero Order Kinetics:

Drug is absorbed at a ***constant amount***

e.g. 100mg → 80mg → 60mg → 40mg → 20mg → 0mg

[drug is absorbed at 20mg each time]

Example: controlled released drugs, depot injection / long acting injection

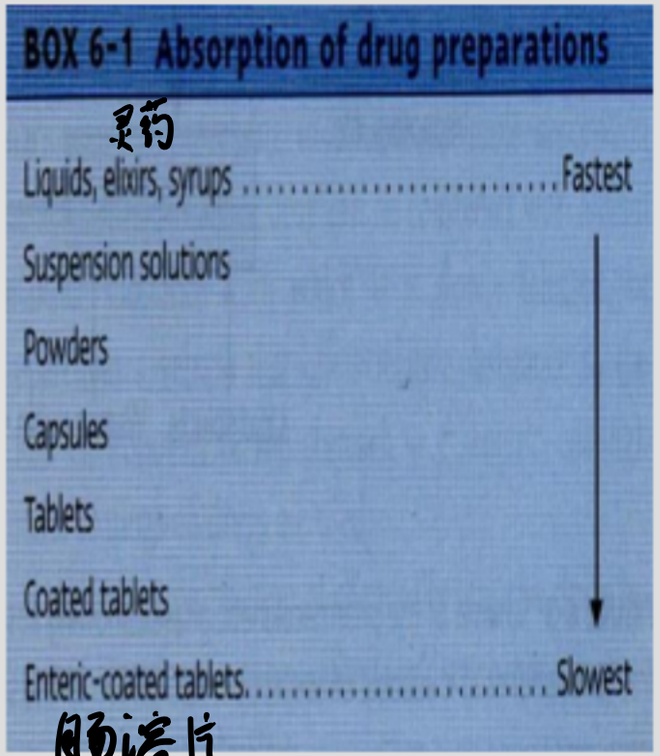
slow-release form of medication designed to dissolve or release into body over weeks, months or longer

ABSORPTION

Factors affecting Absorption:

1. **Route of administration (e.g. IV versus oral)**

2. **Formulation of the drug**



3. **Blood Flow, blood supply**

- Increased blood flow to drug administration site increases absorption
- Decreased blood flow to drug administration site decreases absorption (e.g shock state decreases blood flow to GI)
- Sublingual tablets have a rapid effect due to rapid absorption (rich blood supply underneath the tongue)

4. **Gastrointestinal motility**

- Faster gastric emptying speeds up the rate at which a drug reaches the small intestine which increases the rate of absorption
- Infants, children and elderly have slower gastric emptying rates
- Certain drugs affect gastric motility (e.g metoclopramide ~~reduces~~ increases gastric motility)

ABSORPTION

Factors affecting Absorption (cont.):

5. **Surface area**

- Good absorption at the small intestines → enormous absorptive surface area provided by villi and microvilli
- Larger body surface area, thinner skin (infants, children) → greater absorption through skin

chelating agent grab harmful heavy metals or excess minerals from bloodstream

6. **Presence of food**

- Chelation of drugs (e.g. quinolones with milk) reduces drug absorption
- Presence of food (especially fatty food) slows down gastric emptying

(chelate complex much larger & bulkier than standalone drug molecule)
(chelate complex often highly charged or insoluble, preventing it from diffusing across lipid membranes of intestinal wall.

7. **p-Glycoproteins**

- a reverse transporter located at the intestinal wall
- actively pumps drug out of gutwall cells back into the gut lumen.
- Inhibition of P-glycoprotein can happen eg, by grapefruit juice which substantially increased drug absorption.

ABSORPTION

Factors Affecting Absorption (cont):

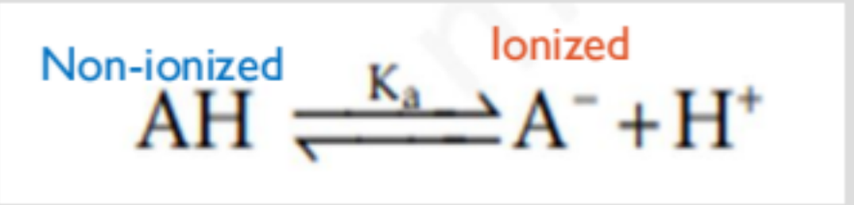
7. Physicochemical properties of the drug

- pH of the environment and drug ionization
- Small and unionized drugs are almost completely absorbed
- Large ionized drugs are difficult to absorb

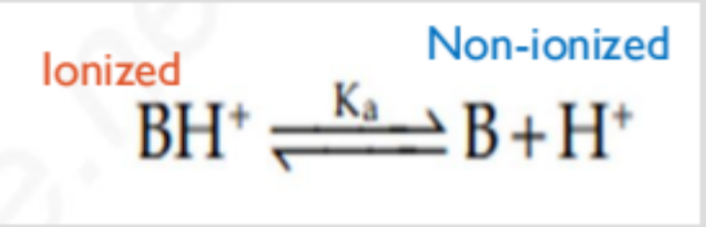
→ Drugs in an ionized form has low lipid solubility, thus harder to permeate cell membranes.

- Drugs can exist in both ionized and non-ionized form, ionization depends on the pH of the environment and pKa of a drug.
- pKa is the pH at which the drug is 50% ionized

Acidic drug:



Basic drug:



Acidic drugs more easily absorbed in acidic environments.

Basic drugs are more easily absorbed in basic environments.

DISTRIBUTION

- **Distribution is the movement of drugs from the circulation to the tissues and organs.**

- Drugs are distributed first to those areas with extensive blood supply.
✓ Areas of rapid distribution include the heart, liver, kidneys, and brain.
✓ Areas of slower distribution include muscle, skin, and fat.

- In pharmacokinetics, we use **Volume of Distribution, Vd** to describe where the drug may be distributed.
✓ If a drug has small Vd, it means it remains mostly in the blood circulation
✓ If a drug has a large Vd, it means it is mostly distributed in the peripheral tissues (muscle, skin, fat).

- Factors influencing distribution of drug:
✓ blood supply
✓ drug solubility
✓ physiologic barriers
✓ drug protein binding properties
✓ patient factors

DISTRIBUTION

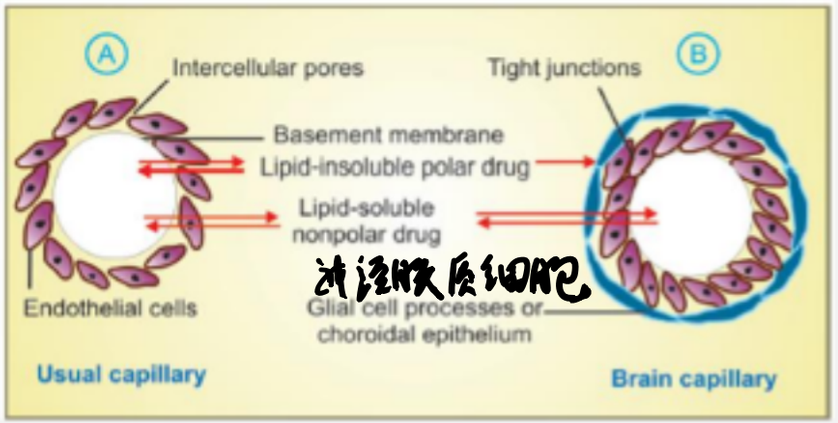
Factors that affect Drug Distribution:

1. **Blood supply**:

✓ Areas of rapid distribution include the heart, liver, kidneys, and brain.

✓ Areas of slower distribution include muscle, skin, fat and bone

In conditions of shock (e.g. septic shock, cardiogenic shock), blood flow is reduced to the organs. Drugs may not be delivered to target sites.



2. **Drug solubility:**

✓ Water-soluble (hydrophilic)drugs will have high blood concentrations and small Vd

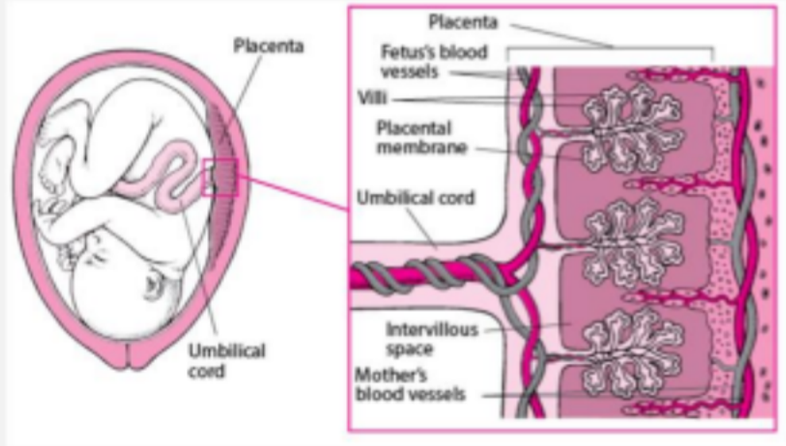
✓ Fat-soluble (lipophilic) drugs will have low blood concentrations and high Vd

3. **Physiologic Barriers:**

✓ Blood brain barrier

- Many drugs do not cross blood brain barrier. Lipophilic drugs crosses more easily.
- However, permeability can increase if the meninges is inflamed / brain trauma / tumor.

✓ Placental transfer:



- Large polar molecules crosses the placenta slower

- Lipophilic drugs pass more readily

- If drug crosses, there's a possibility of adverse effects to fetus

DISTRIBUTION

Factors that affect Drug Distribution (Cont.):

4. Drug Protein Binding

- Plasma protein mostly associated with drug binding is with **albumin.**
- Albumin binds many acidic drugs e.g. phenytoin, warfarin & NSAIDs
- A drug that binds to albumin is a bounded form which is inactive. It cannot cross the blood plasma to get distributed to other compartments.
- A free drug is not binded to albumin is released for distribution and is active.
- **Protein binding is saturable.** If more drug is added, more free drug is available and this can cause erratic / irregular drug action.
- Also one drug can displace the protein binding of another drug, making it a free form which increases it's activity e.g. warfarin & NSAIDs

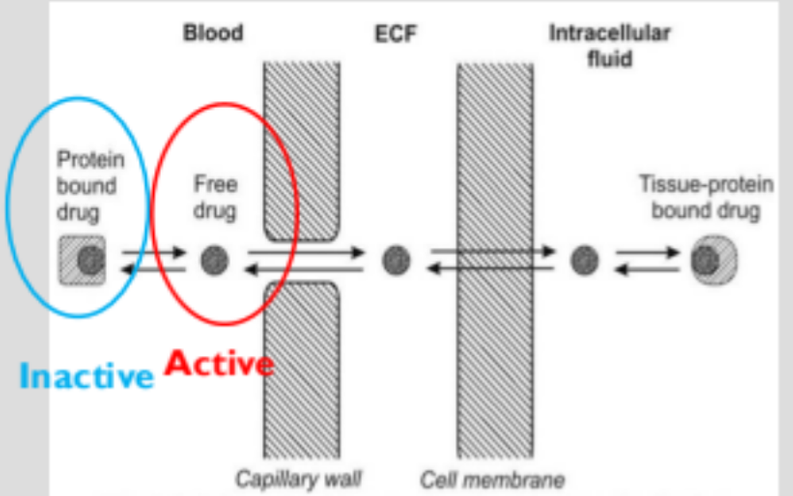


Fig. 3.2. Schematic of the steps involved in drug distribution

DISTRIBUTION

Factors that affect Drug Distribution (Cont.):

5. Patient factors:

- **Neonate & Pediatric:**

✓ greater total body water → higher Vd for hydrophilic drugs
✓ Protein binding capacity is decreased (immature liver)
✓ Developing blood-brain barrier allows more drugs to enter the brain → higher risk to CNS toxicity

- **Older Adult:**

✓ Decrease in total body water and muscle mass.
✓ Body fat may increase and result in a longer duration of action for lipid soluble medications.
✓ Serum albumin often also decreases, resulting in more active free drug within the body → older adult patients may require lower levels of medication

- In CKD and Cirrhosis, albumin is reduced, less protein binding. Increases free drug, can lead to toxicities.

END OF PHARMACOKINETIC I